

# Social Determinants in Idaho

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## 1 Introduction

### 1.1 Background

There is a growing proportion of parents of young children, who hesitate or refuse to vaccinate their children in the United States. Several immunizations are state-mandated for children in public schools. Among those are Tetanus, Diphteria, Pertussis, Hepatitis B, Measles, Mumps, Rubella, Varicella. Although all states allow children to be exempted from immunization for medical reasons, some states are more stringent than others in allowing exemptions to immunization for philosophical or religious reasons. Under the leadership of the current republican government, federal guidelines for recommended childhood immunizations are shifting and there are consequences in the stringency of immunization mandates for children attending public schools at the level of states. In a previous analysis, we found that the highest level of immunization resistance in 2024 is found in the state of Idaho reaching 15%.

## 1.2 Project Question

Focusing on the state of **Idaho**, we will try to find correlations between political affiliation and social determinants such as education levels, median income, poverty rates, population density, language isolation by comparing data at the level of counties, based on the hypothesis that immunization resistance is linked to party affiliation (Iasulaitis (2025)) and/or lack of education (Vardavas (2023)). I do not have access to immunization resistance data at the level of counties even though I requested it from the Idaho Department of Health. Lacking this crucial piece of data, I will look at possible correlations between social determinants, e.g. median income, poverty levels, language isolation, population density, education levels, rates of voter registration by political parties, and also look if correlations can be seen between education levels and median income (Edwards (2024)). The data available for Idaho residents are social determinants values for each county. Because we do not have access to information on individual Idaho residents, we work with the crude approximation that counties represent a concentration of subjects with a given value of a trait (variable) and we will compare it with values observed for other traits (variables). This is not a good approach but is the only available given the data at hand. For the variable “population count” per county, we make the approximation that each county is of the same surface, hence the population count is a representation of density of population. The value for the median income is not independent of the % of population at the poverty level, but the two variables report on different statistics for each county.

## 1.3 Data Sets Overview

We use voter registration data as of January 2025 from the website <https://voteidaho.gov/data-and-dashboards/voter-registration-totals/>. The data for Idaho counties poverty rates, median income, language isolation, education levels, population are collected for the year 2019-2023 and are available at <https://hdpulse.nimhd.nih.gov>.

```
# Set working directory
# C:\Users\alexm\REPOS\CRANE\VaccinationHesitancy
setwd("C:/Users/alexm/REPOS/CRANE/VaccinationHesitancy/Analysis")
```

## 2 Data for Social Determinants in Idaho

We found publicly available data on Social Determinants in Idaho (Education levels, Poverty Levels, Median Income, Language Isolation Levels, Voter Affiliations) per county and for the years 2019 to 2023.

```
# loading the data from https://hdpulse.nimhd.nih.gov and Idaho Board of Election

data_poverty <- read.csv("../data/IdahoCounties_Poverty.csv", skip = 4)
data_education <- read.csv("../data/IdahoCounties_Education.csv", skip = 5)
data_medianincome <- read.csv("../data/IdahoCounties_MedianIncome.csv", skip = 4)
data_languageisolation <- read.csv("../data/IdahoCounties_LanguageIsolation.csv", skip = 4)
data_population <- read.csv("../data/IdahoCounties_Population.csv")
data_voters <- read.csv("../data/2025IdahoVoters.csv", skip = 1)

# change class of Voter Counts to Integer
data_voters$CON <- as.integer(
  gsub(",", "", data_voters$CON))
data_voters$DEM <- as.integer(
  gsub(",", "", data_voters$DEM))
data_voters$LIB <- as.integer(
  gsub(",", "", data_voters$LIB))
```

```
data_voters$REP <- as.integer(
  gsub(",", "", data_voters$REP))
data_voters$UNA <- as.integer(
  gsub(",", "", data_voters$UNA))
data_voters$Total <- as.integer(
  gsub(",", "", data_voters$Total))
```

## 2.1 Voter Affiliations in Fringe Parties in Idaho

We first look at the distribution of Idaho voter registrations in fringe political parties, such as Libertarian and Constitution, which are known to embrace “extreme” political views. We combine the percentage of voters for the Libertarian Party with those for the Constitution Party and call it the percentage of voters for Fringe Parties.

```
# sum voter registration for fringe parties
data_voters$FRINGE.PERCENT <- data_voters$CON.PERCENT + data_voters$LIB.PERCENT
```

The following histogram looks at “Fringe” Party voters (Libertarian and Constitution affiliated voters are summed). The values of “Fringe” Party voters are pooled in 10 bins for clarity. From the distribution (Fig. 1), we can conclude that the rate of Fringe Party Voters does not exceed 2%, hence it is a low subset of the Idaho population, but probably an under representation as one of the tenet of the Libertarian party is to discourage any participation in civic duties such as voting. But because the numbers are so low, we will not use this variable for our analysis. In contrast, the values espoused by the % of Republican-registered voters span a large spectrum (Fig. 2), hence we will use that variable in our analysis.

```
library(patchwork)
library(ggplot2)
hist_fringe <- ggplot(data_voters, aes(x = FRINGE.PERCENT)) +
  geom_histogram(bins = 10,
    fill = "lightblue",
    color = "black") +
  labs(caption = "",
    x = "% Registered as Libertarian or Constitution",
    y = "number of counties")

# county with Max Fringe Voters - which county has the highest value?
max(data_voters$FRINGE.PERCENT)
data_voters$COUNTY[which.max(data_voters$FRINGE.PERCENT)]
data_voters$FIPS[which.max(data_voters$FRINGE.PERCENT)]

hist_dem <- ggplot(data_voters, aes(x = DEM.PERCENT)) +
  geom_histogram(bins = 10,
    fill = "lightblue",
    color = "black") +
  labs(caption = "",
    x = "% Registered as Democrat",
    y = "number of counties")

hist_rep <- ggplot(data_voters, aes(x = REP.PERCENT)) +
  geom_histogram(bins = 10,
    fill = "lightblue",
    color = "black") +
```

```

  labs(caption = "",
        x = "% Registered as Republican",
        y = "number of counties")

hist_ind <- ggplot(data_voters, aes(x = UNA.PERCENT)) +
  geom_histogram(bins = 10,
                 fill = "lightblue",
                 color = "black") +
  labs(caption = "",
        x = "% Registered as Independent",
        y = "number of counties")

# county with Max Republican registered Voters - which county has the highest value?
max(data_voters$REP.PERCENT)
data_voters$COUNTY[which.max(data_voters$REP.PERCENT)]
data_voters$FIPS[which.max(data_voters$REP.PERCENT)]

Fig1 <- (hist_dem + hist_rep) / (hist_ind + hist_fringe) +
  plot_annotation(caption = "Figure 1: Distribution of counties per % voter affiliation",
                  theme = theme(plot.caption = element_text(size=12, hjust=0.5))
  )

# save plot to a output file
ggsave(
  filename = "../output/Counties_n_VoterRegistration.png",
  plot = Fig1,
  width = 8.5,
  height = 10,
  dpi = 300
)

```

## 2.2 Merging of Data

We merge the data available for Idaho residents based on variable “FIPS”, which is a numerical code assigned to each county.

```

library(dplyr)

data <- data_poverty %>%
  full_join(data_education, by = "FIPS") %>%
  full_join(data_medianincome, by = "FIPS") %>%
  full_join(data_languageisolation, by = "FIPS") %>%
  full_join(data_population, by = "FIPS") %>%
  full_join(data_voters, by = "FIPS")

# rename columns
colnames(data)[1] <- "county"
colnames(data)[3] <- "below.poverty.percent"
colnames(data)[7] <- "below.education.percent"
colnames(data)[11] <- "median.income"

```

```

colnames(data)[14] <- "language.isolation.percent"
colnames(data)[19] <- "population.age.18.39"

head(data$median.income)

# change class of median income from character to integer
data$median.income <- as.integer(
  gsub(",", "", data$median.income)
)

# write csv file for merged date
write.csv(data, "../data/social_Idaho.csv")

```

### 3 Exploration of relationships between variables assuming Linearity

#### 3.1 Data Exploration for Population & Voter Registrations

```

library(patchwork)
library(scales)
library(ggplot2)
hist_pop <- ggplot(data, aes(x = population.age.18.39)) +
  geom_histogram(bins = 20,
    fill = "lightblue",
    color = "black") +
  labs(x = "population age 18-39",
    y = "number of counties")

Fig2 <- hist_pop +
  plot_annotation(caption = "Figure 2: Distribution of counties with population age 18-39",
    plot.caption = element_text(size=12, hjust=0.5)
  )

# save plot to a output file
ggsave(
  filename = "../output/Counties_n_Population.png",
  plot = Fig2,
  width = 8.5,
  height = 10,
  dpi = 300
)

# county with Max Fringe Voters - which county has the highest value?
max(data$population.age.18.39)
data$COUNTY[which.max(data$population.age.18.39)]
data$FIPS[which.max(data$population.age.18.39)]

# visualisation of population numbers and republican registrations %
library(ggplot2)

```

```

density_vs_rep <- ggplot(data, aes(x = population.age.18.39, y = REP.PERCENT)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_x_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "Population 18-39",
    y = "Republican registrations (%)" ) +
  theme_minimal()

# visualisation of population numbers and democrat registrations %
library(ggplot2)
density_vs_dem <- ggplot(data, aes(x = population.age.18.39, y = DEM.PERCENT)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_x_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "Population 18-39",
    y = "Democrat registrations (%)" ) +
  theme_minimal()

# visualisation of population numbers and independant registrations %
library(ggplot2)
density_vs_ind <- ggplot(data, aes(x = population.age.18.39, y = UNA.PERCENT)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_x_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "Population 18-39",
    y = "Independent registrations (%)" ) +
  theme_minimal()

# visualisation of population numbers and fringe registrations %
library(ggplot2)
density_vs_fringe <- ggplot(data, aes(x = population.age.18.39, y = FRINGE.PERCENT)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_x_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "Population 18-39",
    y = "Fringe registrations (%)" ) +
  theme_minimal()

Fig3 <- (density_vs_rep + density_vs_dem) /
  (density_vs_ind + density_vs_fringe) +
  plot_annotation(caption = "Figure 3: Voter registration vs county population density",
    theme = theme(plot.caption = element_text(size=12, hjust=0.5))
  )

# save plot to a output file
ggsave(

```

```

filename = "../output/Population_n_VoterRegistration.png",
plot = Fig3,
width = 8.5,
height = 10,
dpi =300
)

```

### 3.2 Data Exploration for Median Income and Education Level, Voter Affiliation, Language Isolation, Below Poverty Level

```

str(data)

library(patchwork)
library(scales)

# visualisation of median.income and below.education.percent
library(ggplot2)
edu_vs_income <- ggplot(data, aes(x = below.education.percent, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "below education (%)",
    y = "median income ($)" ) +
  theme_minimal()

# visualisation of median income & language isolation
library(ggplot2)
lang_vs_income <- ggplot(data, aes(x = language.isolation.percent, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "language isolation (%)",
    y = "median income ($)" ) +
  theme_minimal()

# visualisation of median.income and below.education.percent
library(ggplot2)
poverty_vs_income <- ggplot(data, aes(x = below.poverty.percent, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "below poverty level (%)",
    y = "median income ($)" ) +
  theme_minimal()

```

```

# visualisation of median.income and republican registrations %
library(ggplot2)
rep_vs_income <- ggplot(data, aes(x = REP.PERCENT, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "republican registrations (%)",
    y = "median income ($)" ) +
  theme_minimal()

# visualisation of median.income and democrat registrations %
library(ggplot2)
dem_vs_income <- ggplot(data, aes(x = DEM.PERCENT, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "democrat registrations (%)",
    y = "median income ($)" ) +
  theme_minimal()

# visualisation of median.income and independent registrations %
library(ggplot2)
ind_vs_income <- ggplot(data, aes(x = UNA.PERCENT, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "independent registrations (%)",
    y = "median income ($)" ) +
  theme_minimal()

# visualisation of median.income and fringe registrations %
library(ggplot2)
fringe_vs_income <- ggplot(data, aes(x = FRINGE.PERCENT, y = median.income)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  geom_smooth(method=lm) +
  labs(
    x = "libertarian & constitution registrations (%)",
    y = "median income ($)" ) +
  theme_minimal()

Fig4 <- (edu_vs_income / lang_vs_income / poverty_vs_income) +
  plot_annotation(caption = "Figure 4: County median income vs social determinants",
    theme = theme (plot.caption = element_text(size=12, hjust=0.5)
    ))

# save plot to a output file
ggsave(

```



```

filename = "../output/Income_n_EducationPovertyLanguage.png",
plot = Fig4,
width = 8.5,
height = 10,
dpi =300
)

Fig5 <- (rep_vs_income + dem_vs_income) /
(ind_vs_income + fringe_vs_income) +
  plot_annotation(caption = "Figure 5: County median income vs voter registration",
    theme = theme (plot.caption = element_text(size=12, hjust=0.5)
    ))

# save plot to a output file
ggsave(
  filename = "../output/Income_n_VoterRegistration.png",
  plot = Fig5,
  width = 8.5,
  height = 10,
  dpi =300
)

```

Several variables seem consistent with linear correlations pair-wise.

## 4 Model building

We will build a model to analyze correlations between Median Income and other social determinants. This does not address directly the original question. We here ask whether counties with higher median income have lower republican voter percentages, higher density of population, lower language isolation percentages, lower poverty rates, and lower below education percentages. The assumption of linearity was more or less established by the pair-wise scatter plots displayed above.

```

# fit model

mod= lm( median.income ~ REP.PERCENT + language.isolation.percent + population.age.18.39 + below.poverty.percent + below.education.percent,
  data = data)

summary(mod)

##
## Call:
## lm(formula = median.income ~ REP.PERCENT + language.isolation.percent +
##      population.age.18.39 + below.poverty.percent + below.education.percent,
##      data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13395.1  -4655.9  -332.9   4432.2  18255.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)

```

```
## (Intercept)          1.145e+05  7.776e+03  14.728 < 2e-16 ***
## REP.PERCENT          -3.147e+02  1.044e+02  -3.015  0.00456 **
## language.isolation.percent -5.660e+02  3.169e+02  -1.786  0.08208 .
## population.age.18.39    1.030e-01  4.649e-02   2.216  0.03278 *
## below.poverty.percent  -1.616e+03  3.659e+02  -4.416  8.08e-05 ***
## below.education.percent -5.853e+01  3.828e+02  -0.153  0.87930
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7238 on 38 degrees of freedom
## Multiple R-squared:  0.6165, Adjusted R-squared:  0.5661
## F-statistic: 12.22 on 5 and 38 DF,  p-value: 4.283e-07
```

## 4.1 Verification of the assumption of absence of collinearity

```
library(car)

# VIF
vif(mod)

##              REP.PERCENT language.isolation.percent
##              1.194043              1.311192
##      population.age.18.39      below.poverty.percent
##              1.184740              1.058487
##      below.education.percent
##              1.317643
```

All VIF values are below 1.2, hence collinearity is likely not present between the explanatory variables.

## 5 Model Selection

```
mod1= lm( median.income ~ below.poverty.percent , data=data)

summary(mod1)

##
## Call:
## lm(formula = median.income ~ below.poverty.percent, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15194  -6184  -2010   6096  24803
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      96474.0      3921.9   24.60 < 2e-16 ***
## below.poverty.percent -2063.8       443.8   -4.65 3.29e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9033 on 42 degrees of freedom
## Multiple R-squared:  0.3398, Adjusted R-squared:  0.3241
## F-statistic: 21.62 on 1 and 42 DF,  p-value: 3.291e-05
```

```
mod2= lm( median.income ~ below.poverty.percent + REP.PERCENT , data=data)

summary(mod2)
```

```
##
## Call:
## lm(formula = median.income ~ below.poverty.percent + REP.PERCENT,
##     data = data)
##
## Residuals:
```

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -14989.8 | -5341.6 | -892.3 | 5406.1 | 15740.3 |

```
##
## Coefficients:
```

|                       | Estimate | Std. Error | t value | Pr(> t )     |
|-----------------------|----------|------------|---------|--------------|
| (Intercept)           | 121467.1 | 7363.8     | 16.495  | < 2e-16 ***  |
| below.poverty.percent | -1762.8  | 393.5      | -4.480  | 5.87e-05 *** |
| REP.PERCENT           | -404.6   | 105.7      | -3.828  | 0.000433 *** |

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7847 on 41 degrees of freedom
## Multiple R-squared:  0.5137, Adjusted R-squared:  0.49
## F-statistic: 21.65 on 2 and 41 DF,  p-value: 3.818e-07
```

```
mod3= lm( median.income ~ below.poverty.percent + REP.PERCENT + population.age.18.39, data=data)

summary(mod3)
```

```
##
## Call:
## lm(formula = median.income ~ below.poverty.percent + REP.PERCENT +
##     population.age.18.39, data = data)
##
## Residuals:
```

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -13137.3 | -4895.2 | -200.4 | 4569.9 | 18282.3 |

```
##
## Coefficients:
```

|                       | Estimate   | Std. Error | t value | Pr(> t )    |
|-----------------------|------------|------------|---------|-------------|
| (Intercept)           | 1.136e+05  | 7.780e+03  | 14.600  | < 2e-16 *** |
| below.poverty.percent | -1.679e+03 | 3.757e+02  | -4.470  | 6.3e-05 *** |
| REP.PERCENT           | -3.191e+02 | 1.070e+02  | -2.983  | 0.00484 **  |
| population.age.18.39  | 1.102e-01  | 4.749e-02  | 2.320   | 0.02550 *   |

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 7458 on 40 degrees of freedom
## Multiple R-squared:  0.5714, Adjusted R-squared:  0.5392
## F-statistic: 17.77 on 3 and 40 DF,  p-value: 1.731e-07
```

```
mod4= lm( median.income ~ below.poverty.percent + REP.PERCENT + population.age.18.39 + language.isolation.percent, data = data)
summary(mod4)
```

```
##
## Call:
## lm(formula = median.income ~ below.poverty.percent + REP.PERCENT +
##     population.age.18.39 + language.isolation.percent, data = data)
##
## Residuals:
```

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -13295.6 | -4696.6 | -300.5 | 4284.3 | 18271.2 |

```
##
## Coefficients:
```

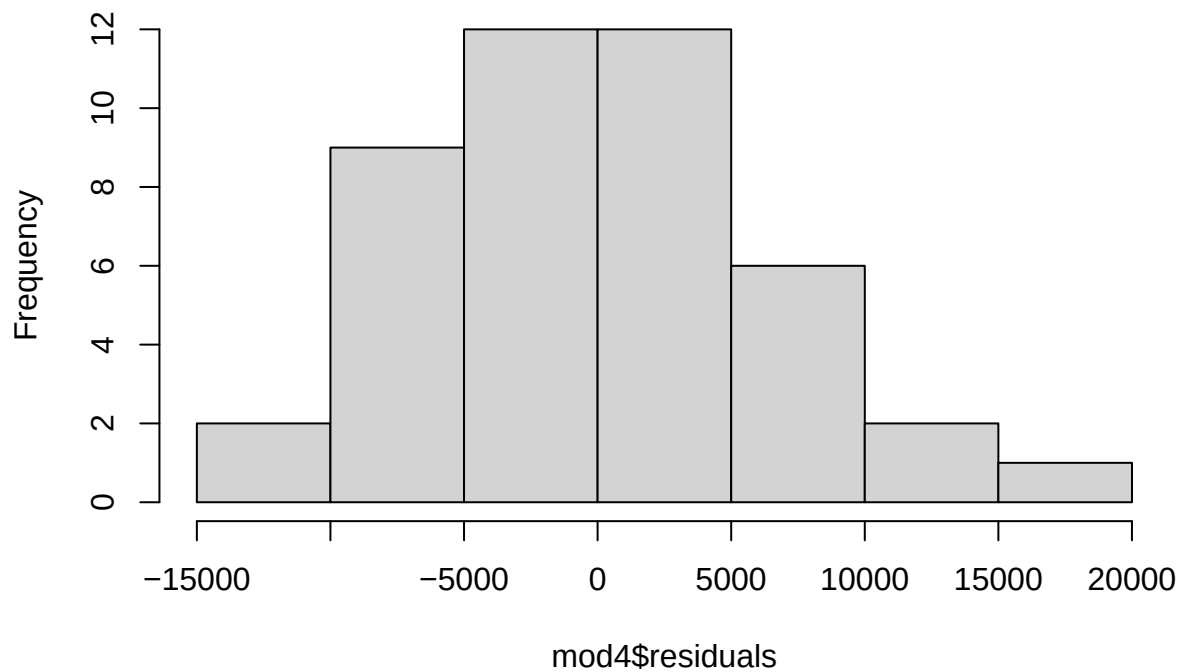
|                            | Estimate   | Std. Error | t value | Pr(> t )     |
|----------------------------|------------|------------|---------|--------------|
| (Intercept)                | 1.142e+05  | 7.461e+03  | 15.311  | < 2e-16 ***  |
| below.poverty.percent      | -1.617e+03 | 3.612e+02  | -4.475  | 6.45e-05 *** |
| REP.PERCENT                | -3.131e+02 | 1.025e+02  | -3.053  | 0.00406 **   |
| population.age.18.39       | 1.038e-01  | 4.560e-02  | 2.277   | 0.02837 *    |
| language.isolation.percent | -5.889e+02 | 2.756e+02  | -2.137  | 0.03896 *    |

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7147 on 39 degrees of freedom
## Multiple R-squared:  0.6163, Adjusted R-squared:  0.5769
## F-statistic: 15.66 on 4 and 39 DF,  p-value: 1.006e-07
```

## 5.1 Verification of the assumptions of normality and homoscedasticity of residuals for Model 4

```
hist(mod4$residuals)
```

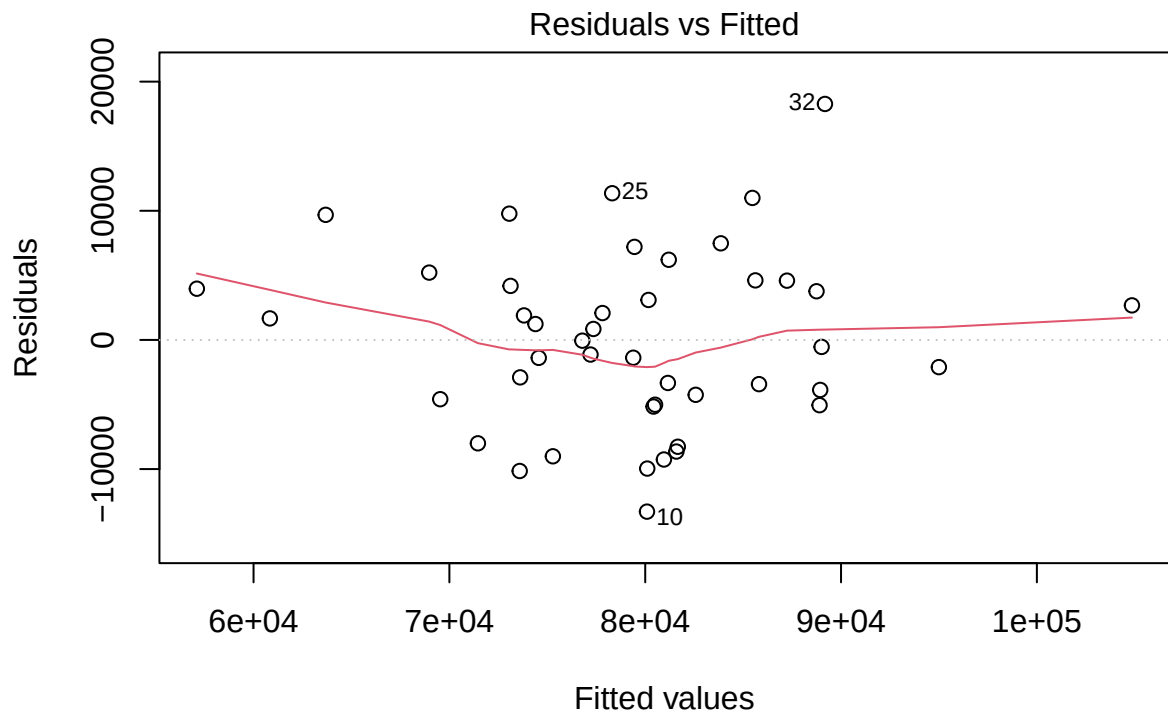
## Histogram of mod4\$residuals



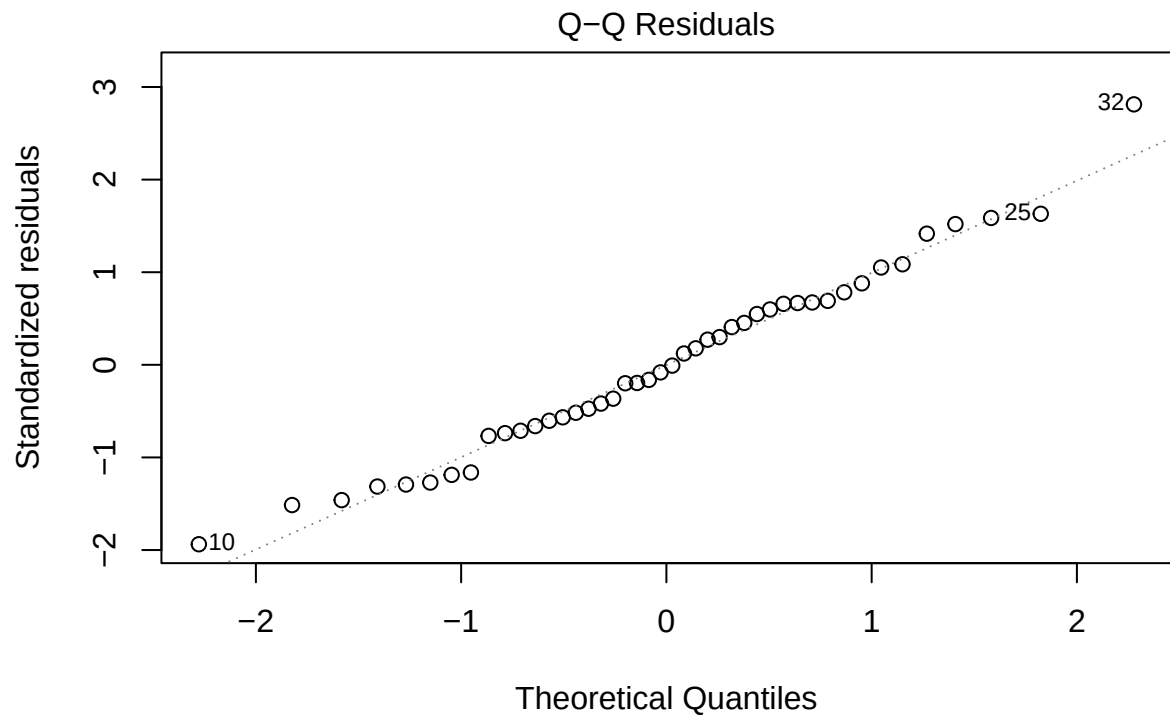
```
shapiro.test(mod4$residuals)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  mod4$residuals  
## W = 0.98476, p-value = 0.8202
```

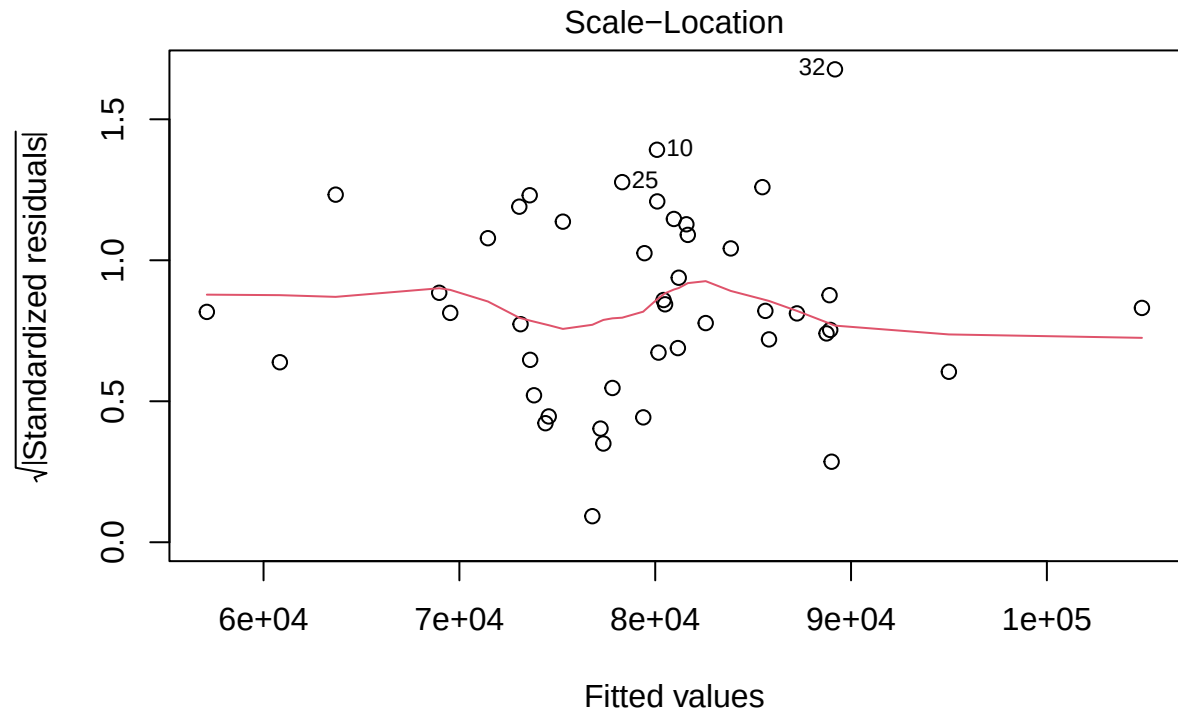
```
plot(mod4)
```



lm(median.income ~ below.poverty.percent + REP.PERCENT + population.age.18. .

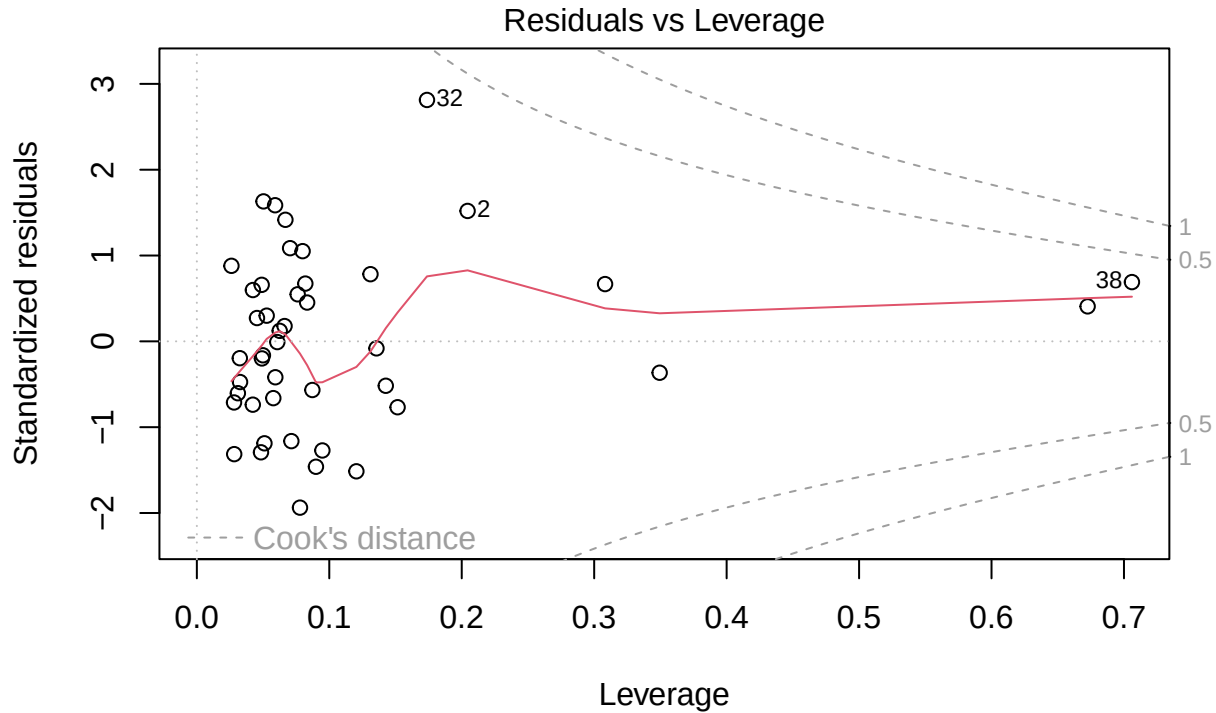


lm(median.income ~ below.poverty.percent + REP.PERCENT + population.age.18. .



lm(median.income ~ below.poverty.percent + REP.PERCENT + population.age.18. .





*# calculate mean and sd of median income variable and population variable*

```
meanI <- mean(data$median.income)
sdI <- sd(data$median.income)
medianP <- median(data$population.age.18.39)
IQRP <- IQR(data$population.age.18.39)
```

The normality of the residuals is visible from the histogram. The Shapiro-Wilk normality test also gives a p-value of 0.82, hence the null hypothesis (absence of normality) can be rejected. The Residuals vs Fitted plot is not quite centered along a straight line at  $y=0$  but we will consider that there is homoscedasticity of the residuals.

## 6 Conclusion

The correlation between median income and below poverty %, republican party affiliation, population age 18-39 and language isolation % appears significant with p values  $<0.05$ .

| Variable               | Effect Size, SD 95% CI | unit   | P-value |
|------------------------|------------------------|--------|---------|
| below poverty (%)      | $-1617 \pm 708$        | \$/%   | 6.5e-05 |
| republican (%)         | $-313 \pm 200$         | \$/%   | 0.00406 |
| population 18-39       | $104 \pm 89$           | \$/100 | 0.02837 |
| language isolation (%) | $-589 \pm 540$         | \$/%   | 0.03896 |

The adjusted R-squared value is 0.57.

The effect sizes when relating median income to % numbers are large because of the large scale differential (the average for the median income is  $7.9372932 \times 10^4$  with Standard Deviation  $1.0987614 \times 10^4$ ), while percentages are between 0 and 100. The effect size when relating median income to populations numbers is small because the county median population is 3594 and the IQR is 8569.75, hence in a more similar scale than the median income. Note that the population variable distribution is not normal- there are many scarcely populated counties and one very urban one, and not many in between. Hence this variable lacks data points for the larger range. This may skew the calculation of the effect size.

The statistical analysis reports on the effect sizes on Idaho county Median Income of the % of population at poverty level, the % of county resident affiliated as Republicans, the population density, the % population suffering language isolation. Not surprisingly, the % of poverty level is inversely related to the county median income and so are the % of the population experiencing language isolation the % of the population affiliated to the Republican party. In contrast, median income correlates positively with population density. In other words, counties with low density of population and higher poverty % and higher language isolation % and higher Republican political affiliation %, have lower median income.

## 7 AI Use Disclosure Statement

This analysis did not rely on A.I.

## 8 GitHub link

<https://github.com/ac292/CRANE>

## Literature Cited

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